

Lecture Scribe: Linear Transformations

L20

 PDF

Task Instructions: Using the same structure as the example provided above, generate a comprehensive lecture scribe based on the attached context for Lecture 20(L20.pdf). This scribe is authored by Vansh Lilani (ID: AU2320146). It is intended to serve as a primary exam-oriented reference material for a reading-based/closed-notes exam.

Requirements:

Faithfulness & Detail: Include all definitions, notation, and assumptions exactly as presented. Every formula, theorem statement, and worked example from the lecture must be included in full.

Step-by-Step Explanations: For all proofs and derivations, provide the complete step-by-step logic shown in the slides. Do not summarize; ensure the explanation covers the transition between each mathematical step to ensure exam readiness.

Scope: Stick strictly to the provided PDF. Do not invent new content or add external "intuition" not present in the lecture materials.

Correctness & Tone: Maintain a formal academic tone. Prioritize mathematical precision and clarity in the layout of formulas.

Structure: Adhere strictly to the organizational pattern (headings, lists, and layout) demonstrated in the "Example Lecture Scribe" provided above.



CSE400: Fundamentals of Probability in Computing Lecture 20: Linear Transformations and Expectations of Multiple RVs Course Instructor: Dhaval Patel, PhD

1. Statistical Representation of Random Vectors

For multiple random variables $X_1, X_2, \dots, X_N$, we utilize a compact vector representation to manage their interdependencies.

1.1 Random Vector Definition

A **random vector** $\underline{X}$ is a column vector containing N random variables:

$$\underline{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix} \quad [\text{cite: 3}]$$

1.2 Mean Vector

The **mean vector** $\underline{\mu}_X$ represents the average value of the random vector, where each element corresponds to the expected value of an individual random variable:

$$\underline{\mu}_X = E[\underline{X}] = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix} \quad [\text{cite: 4, 8}]$$

1.3 Correlation Matrix

The **correlation matrix** R_{XX} captures the correlation between multiple random variables.

- **Definition:** $R_{XX} = E[\underline{X}\underline{X}^T]$.
- **Element Representation:** For $i, j \leq N$, the element $R_{XX}(i, j) = E[X_i X_j]$.
- **Matrix Structure:**

$$R_{XX} = \begin{bmatrix} E[X_1 X_1] & E[X_1 X_2] & \dots & E[X_1 X_N] \\ E[X_2 X_1] & E[X_2 X_2] & \dots & E[X_2 X_N] \\ \vdots & \vdots & \ddots & \vdots \\ E[X_N X_1] & E[X_N X_2] & \dots & E[X_N X_N] \end{bmatrix}_{N \times N} \quad [\text{cite: 8}]$$

- **Property:** R_{XX} is a symmetric matrix.

1.4 Covariance Matrix

The **covariance matrix** C_{XX} serves as a "relationship map" for the random vector.

- **Definition:** $C_{XX} = E[(\underline{X} - \underline{\mu}_X)(\underline{X} - \underline{\mu}_X)^T]$.
- **Practical Structure:**

$$C_{XX} = \begin{bmatrix} Var(X_1) & Cov(X_1, X_2) & \dots \\ Cov(X_2, X_1) & Var(X_2) & \dots \\ \vdots & \vdots & \ddots \end{bmatrix} \text{ [cite: 10]}$$

- **Property:** Due to the fact that $Cov(X_i, X_j) = Cov(X_j, X_i)$, C_{XX} is symmetric.
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2. Linear Transformation of Random Vectors

We consider a linear model where an input random vector $\underline{X}$ is transformed into an output vector $\underline{Y}$.

2.1 The Linear Model

The transformation is defined as:

$$\underline{Y} = \underline{A}\underline{X} + \underline{b} \text{ [cite: 11]}$$

- $\underline{A}$: Transformation matrix (scales and rotates data).
- $\underline{b}$: Constant vector (shifts/translates the center).
- $\underline{X}$: Input random vector.

For N -dimensional vectors, this expands to:

$$Y_1 = a_{11}X_1 + a_{12}X_2 + \dots + a_{1N}X_N + b_1$$

$$\vdots$$

$$Y_N = a_{N1}X_1 + a_{N2}X_2 + \dots + a_{NN}X_N + b_N$$

3. Properties of Transformed Vectors

When $\underline{Y} = \underline{A}\underline{X} + \underline{b}$, the statistical properties of $\underline{Y}$ are derived as follows:

3.1 Mean of Transformed Vector ($\underline{\mu}_Y$)

- **Derivation:**

1. Substitute $\underline{Y} = \underline{A}\underline{X} + \underline{b}$ into the expectation: $\underline{\mu}_Y = E[\underline{A}\underline{X} + \underline{b}]$.
2. Apply linearity of expectation: $E[\underline{A}\underline{X}] + E[\underline{b}]$.

- **Result:** $\underline{\mu}_Y = \underline{A}\underline{\mu}_X + \underline{b}$.

3.2 Covariance of Transformed Vector (C_{YY})

- **Key Observation:** Deviation from the mean is independent of shift $\underline{b}$:

$$\underline{Y} - \underline{\mu}_Y = (\underline{A}\underline{X} + \underline{b}) - (\underline{A}\underline{\mu}_X + \underline{b}) = \underline{A}(\underline{X} - \underline{\mu}_X)$$

- **Derivation:**

1. $C_{YY} = E[(\underline{Y} - \underline{\mu}_Y)(\underline{Y} - \underline{\mu}_Y)^T]$.
2. Substitute the observation: $C_{YY} = E[\underline{A}(\underline{X} - \underline{\mu}_X)(\underline{X} - \underline{\mu}_X)^T \underline{A}^T]$.

- **Simplified Result:** $C_{YY} = \underline{A}C_{XX}\underline{A}^T$.

3.3 Correlation of Transformed Vector (R_{YY})

- **Derivation:**

1. $R_{YY} = E[\underline{Y}\underline{Y}^T] = E[(\underline{A}\underline{X} + \underline{b})(\underline{A}\underline{X} + \underline{b})^T]$.
2. Expansion of terms: $R_{YY} = E[\underline{A}\underline{X}\underline{X}^T \underline{A}^T + \underline{A}\underline{X}\underline{b}^T + \underline{b}\underline{X}^T \underline{A}^T + \underline{b}\underline{b}^T]$.
3. Applying expectation: $R_{YY} = \underline{A}E[\underline{X}\underline{X}^T]\underline{A}^T + \underline{A}E[\underline{X}]\underline{b}^T + \underline{b}E[\underline{X}^T]\underline{A}^T + \underline{b}\underline{b}^T$.

- **Final Result:** $R_{YY} = \underline{A}R_{XX}\underline{A}^T + \underline{A}\underline{\mu}_X\underline{b}^T + \underline{b}\underline{\mu}_X^T \underline{A}^T + \underline{b}\underline{b}^T$.

4. Numerical Example

Given: $\underline{Y} = \underline{A}\underline{X}$ where:

$$\underline{\mu}_X = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \underline{A} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix}, R_{XX} = \begin{bmatrix} 13 & 2 & 3 \\ 2 & 10 & 3 \\ 3 & 3 & 10 \end{bmatrix}$$

Solution (a): Mean Vector of $\underline{Y}$

Using $\underline{\mu}_Y = \underline{A}\underline{\mu}_X$:

$$\underline{\mu}_Y = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} (1)(2) + (0)(-1) + (-1)(1) \\ (0)(2) + (-1)(-1) + (1)(1) \\ (-1)(2) + (1)(-1) + (0)(1) \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$$

5. Case Study: Smart City Traffic & Weather (Ahmedabad)

We model a city's congestion using five random variables: Rainfall, Temperature, Traffic Volume, Speed, and Pollution.

- **Assumption:** The entire system behaves as a **Joint Gaussian Distribution**, defined by its Mean (typical conditions) and Covariance (interdependencies).
- **Inferences from Covariance Matrix:**
 - $\uparrow$ Rainfall $\implies$ $\uparrow$ Traffic (Congestion).
 - $\uparrow$ Rainfall $\implies$ $\downarrow$ Speed.
 - $\uparrow$ Traffic $\implies$ $\uparrow$ Pollution.
 - $\uparrow$ Temperature $\implies$ Pollution disperses.
- **Transformation:** A city defined "Congestion Index" is a linear transformation